

Accuracy limitations on Brillouin lidar measurements of temperature and sound speed in the ocean

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There are five mutually dependent variables relevant to Brillouin lidar measurements of temperature and sound speed in the ocean; they are (1) the Brillouin shift, (2) the sound speed, (3) the index of refraction, (4) the temperature, and (5) the salinity. We use three well-known relations to analyze rigorously the interdependence of these five variables. Clearly, a Brillouin shift measurement does not provide a stand-alone determination of temperature or sound speed; one more variable or one more relation must be known. The use of mean values of salinity that have been obtained by an analysis of a large set of historical *in situ* data is considered for this additional relation. © 1997 Optical Society of America

Key words: Brillouin, lidar, temperature, sound speed, salinity, refractive index.

1. Introduction

Accurate remote sensing of the temperature and sound speed structure of the upper-ocean mixed layers is of major importance to the understanding of the physical and biological behavior of the ocean.^{1,2} There are also many other applications; it is, for example, a necessary input to weather forecast and climate change studies.³ There currently exists no instrumentation for remote sensing of these parameters, and only *in situ* measurements have been performed with the required accuracy. However, *in situ* approaches do not allow the rapid, accurate, and real-time range-resolved monitoring that is needed. Lidar remote sensing provides a promising solution. Both Raman and Brillouin backscattering are sensitive to temperature, and Brillouin scattering is related directly to sound speed. Both processes have been considered previously for lidar measurements of the vertical structure of temperature and sound speed in the ocean.²⁻¹² Major limitations of the Raman method relative to the Brillouin technique include: (1) the large Raman frequency shift, which means the transmitted laser field and/or the backscattered field must be in a relatively highly absorbing region of the water spectrum; and (2) the Raman

backscattering spectrum has a much wider frequency distribution than the Brillouin backscattering one. Consequently, the Raman receiver must have a much wider bandwidth and is therefore more susceptible to background and noise.¹¹

Several studies have shown the potential accuracy of lidar measurements of sound speed and temperature in the ocean by measuring the spectral shift of the backscattered Brillouin lines.^{2,7} Leonard and Sweeney¹¹ have stated that, to achieve a temperature accuracy in the range of millidegrees over two or three attenuation lengths, advanced Brillouin methods must be invoked. However, taking into account other relevant factors, we show that, for Brillouin measurements alone (i.e., no additional complementary measurements), the expected accuracy is only some tenths of a degree; this is, however, sufficient for many applications. Previously, technical limitations had prevented the realization of such Brillouin lidar instrumentation. Nevertheless, the required technology, in particular pulsed lasers with a Fourier transform limited bandwidth, are now commercially available,² and a receiver based on an edge technique¹³ that uses absorption lines in Br₂ and I₂ now appears feasible.¹⁴ We emphasize that our objective in this paper is to determine general limits on the accuracy of temperature and sound speed determination using lidar Brillouin shift measurements and not to describe an overall system.

The Brillouin lines consist of two inelastically scattered Lorentzians centered symmetrically at $\pm\nu_B$ with respect to the transmitted laser line; for water the FWHM of the Brillouin lines is approximately 0.5

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GHz and ν_B is 7–8 GHz. For a given incident laser wavelength λ , the relation between the Brillouin shift ν_B and the speed of sound v_s in ocean water is given for 180° backscattering by^{1,2}

$$\nu_B(S, T, p) = \frac{2n(S, T, \lambda)}{\lambda} v_s(S, T, p), \quad (1)$$

where n is the refractive index of water, S is the salinity, T is the temperature, and p is the pressure.

To derive the sound speed from the Brillouin shift measurement, the refractive index n has been frequently assumed to be known.^{1,7,9} A more accurate approach has been considered by Hickman *et al.*² who point out that the uncertainty in our knowledge of the refractive index that is due to the uncertainty in T and S contributes to the error in the measurement of the sound speed. Using an empirical relation for the refractive index as a function of T and S ,¹⁵ and assuming that T is known to $\pm 2^\circ\text{C}$, and that S is known to $\pm 1\%$, they estimate that the refractive-index variability can limit the sound speed accuracy to ~ 0.5 m/s.

To derive the temperature from the Brillouin shift measurement, an empirical equation is generally used to describe the dependence of the sound speed on S , T , and p .¹⁶ Then using the measured Brillouin shift and assuming knowledge of the salinity, the refractive index, and the pressure in Eq. (1), one can obtain the temperature.^{1,9,11}

In this paper we provide a rigorous approach to this interdependence. We identify five variables (ν_B , S , T , v_s , n), and since we have three relations between them, two of these variables must be independent and three must be dependent. Because we assume a measurement of ν_B , we take it as one of the independent variables. We take S as the other independent variable and obtain empirical equations for the three dependent variables T , v_s , and n as a function of the two independent variables ν_B and S . Throughout the discussion we neglect the dependence on pressure since it is a well-known function of depth and can be included later.

Because we have taken S to be the other independent variable, a large set of *in situ* salinity measurements was analyzed to determine the salinity mean value and standard deviation as a function of depth and location. These results enable us to determine a limit on the accuracy in the retrieval of T and v_s from measurements of the Brillouin shift when historical data are used for the salinity mean value and standard deviation.

The relations between ν_B , S , T , v_s , and n are examined in Section 2. The limits on the accuracy of measurements of T and v_s are discussed in Section 3; effects that are due to the estimates of the variability in S and to the errors in the measurement of ν_B are included.

Table 1. Coefficients in the Empirical Expression for the Sound Speed $v_s(S, T)$ as quoted by Del Grosso^a

$c_0 = 1402.392$	$c_1 = 5.01109398873,$
$c_2 = -0.0550946843172$	$c_3 = 0.00022153596924,$
$c_4 = 1.32952290781$	$c_5 = 0.000128955756844,$
$c_6 = -0.0127562783426$	$c_7 = 0.000096840315641.$

^aRef. 16.

2. Relations between Brillouin Shift, Salinity, Temperature, and Sound Speed in the Ocean

The most widely used empirical equation for the dependence of sound speed on salinity, temperature, and pressure in natural water is given by¹⁶

$$v_s(S, T, p) = c_0 + c_1T + c_2T^2 + c_3T^3 + c_4S + c_5S^2 + c_6TS + c_7T^2S + f(S, T, p), \quad (2)$$

where the coefficients c_i are constants and $f(S, T, p)$ represents all the terms containing the pressure. The c_i 's with T in degrees Celsius, S in parts per thousand, p in atmospheres, and v_s in meters per second are given in Table 1, where they are shown with all the digits quoted by Del Grosso.¹⁶ Typical values for the changes in v_s that are due to changes in p , T , and S are

$$\begin{aligned} \left(\frac{\partial v_s}{\partial p} \right)_{S=35\text{‰}, T=10^\circ\text{C}, p=0} &= 0.16 \frac{\text{m/s}}{\text{atm}}, \\ \left(\frac{\partial v_s}{\partial T} \right)_{S=35\text{‰}, T=10^\circ\text{C}, p=0} &= 3.5 \frac{\text{m/s}}{^\circ\text{C}}, \\ \left(\frac{\partial v_s}{\partial S} \right)_{S=35\text{‰}, T=10^\circ\text{C}, p=0} &= 1.22 \frac{\text{m/s}}{\text{‰}}. \end{aligned} \quad (3)$$

We also note that the horizontal variability of pressure at a given depth is negligible. For simplicity, the pressure is assumed to be zero for the present calculations, i.e., $f(S, T, p) = 0$. Generalization to other pressures is straightforward.

An empirical equation for the dependence of the refractive-index n of sea water at atmospheric pressure on temperature, salinity, and wavelength is given by¹⁷

$$n(S, T, \lambda) = n_0 + S(n_1 + n_2T + n_3T^2) + n_4T^2 + \frac{n_5 + n_6S + n_7T}{\lambda} + \frac{n_8}{\lambda^2} + \frac{n_9}{\lambda^3}, \quad (4)$$

where λ is the wavelength in nanometers, T is in degrees Celsius, S is in parts per thousand, and the coefficients n_j are constants, given in Table 2.

Again, the five variables of interest are ν_B , S , T , v_s , and n . Because Eqs. (1), (2), and (4) provide three relations between them, any two of the five can be chosen as independent. Our choices are ν_B and S since the former is the measurement result and the latter is the most stable and well known, as we show in Section 3.A. With these two as independent variables, the other three variables can be expressed as $T(S, \nu_B)$, $v_s(S, \nu_B)$, and $n(S, \nu_B)$.

Table 2. Coefficients in the Empirical Expression for the Refractive-Index $n(S, T, \lambda)^a$

$n_0 = 1.31405$	$n_1 = 1.779 \times 10^{-4}$
$n_2 = -1.05 \times 10^{-6}$	$n_3 = 1.6 \times 10^{-8}$
$n_4 = -2.02 \times 10^{-6}$	$n_5 = 15.868$
$n_6 = 0.01155$	$n_7 = -0.00423$
$n_8 = -4382$	$n_9 = 1.1455 \times 10^6$

^aRef. 17.

To obtain empirical equations for $T(S, \nu_B)$ and $v_s(S, \nu_B)$, we use Eqs. (1), (2), and (4) to generate a table containing $v_s(S, T)$, $n(S, T)$, and $\nu_B(S, T)$ at fixed values of S and T (Table 3). Taking the entries for 45 sets of S , ν_B , and T , we perform a least-squares fit to determine a function $T(S, \nu_B)$ of the form

$$T(S, \nu_B) = t_0 + t_1(\nu_B - 7.5) + t_2(\nu_B - 7.5)^2 + t_3(\nu_B - 7.5)^3 + t_4(\nu_B - 7.5)^6 + S[t_5 + t_6(\nu_B - 7.5) + t_7(\nu_B - 7.5)^2 + t_8(\nu_B - 7.5)^3], \quad (5)$$

where T is in degrees Celsius, ν_B in gigahertz, S in per thousand, and the equation is valid over the ranges of the variables in Table 3. In Eq. (5), the ν_B dependence is taken to be a power series about the point $\nu_B = 7.5$ GHz; the choice of powers minimizes the chi square for this number of terms.

The resulting t_j are given in Table 4. For each (S, ν_B, T) combination listed in Table 3, the difference between $T(S, \nu_B)$ from Eq. (5) and the corresponding value of T from Table 3 is shown in Fig. 1. Numerical evaluations of Eq. (5) are shown in Fig. 2, where T is plotted as a function of S for various values of ν_B (7.4–7.7 GHz in steps of 50 MHz), and in Fig. 3, where T is plotted as a function of ν_B for various values of S (30‰ to 40‰ in steps of 2.5‰).

Similarly, for the sound speed, we use the entries in Table 3 for the sets of S , ν_B , and v_s and perform a least-squares fit to obtain a function $v_s(S, \nu_B)$ of a form, analogous to Eq. (5):

$$v_s(S, \nu_B) = c'_0 + c'_1(\nu_B - 7.5) + c'_2(\nu_B - 7.5)^2 + c'_3(\nu_B - 7.5)^3 + c'_4(\nu_B - 7.5)^5 + S[c'_5 + c'_6(\nu_B - 7.5)^2 + c'_7(\nu_B - 7.5)^3], \quad (6)$$

Table 4. Coefficients in the Empirical Expression for $T(S, \nu_B)$ at 532.57 nm

$t_0 = 23.5$	$t_1 = 65.5$	$t_2 = 75$
$t_3 = 252$	$t_4 = 1100$	$t_5 = -0.402$
$t_6 = -0.287$	$t_7 = -0.902$	$t_8 = -5.5$

where v_s is in meters per second, ν_B is in gigahertz, and S is in parts per thousand.

The resulting c'_j 's are given in Table 5. For each (S, ν_B, v_s) combination listed in Table 3, the difference between $v_s(S, \nu_B)$ from Eq. (6) and the corresponding value of v_s from Table 3 is shown in Fig. 4. Numerical evaluations of Eq. (6) are shown in Fig. 5, where v_s is plotted as a function of S for ν_B from 7.4 to 7.7 GHz in steps of 50 MHz, and in Fig. 6, where v_s is plotted as a function of ν_B for S from 30‰ to 40‰ in steps of 5‰. Equations (5) and (6) together with the coefficients in Tables 4 and 5 are valid for $30‰ \leq S \leq 40‰$ and $7.0 \text{ GHz} < \nu_B < 8.0 \text{ GHz}$.

The dependence of the temperature on the independent variable S for fixed values of the independent variable ν_B is shown in Fig. 2. Quantitatively, from Eq. (5) we find

$$\left(\frac{\partial T}{\partial S}\right)_{\nu_B=7.5\text{GHz}} = t_5 \approx -0.4 \text{ }^\circ\text{C}/\text{‰}. \quad (7)$$

If the uncertainty in the salinity is approximately 0.5‰, the uncertainty in the temperature measurement is ≤ 0.2 °C. In Subsection 3.A we show that such an uncertainty in the salinity is typical in historical ocean data at a given depth and location.

The dependence of T on ν_B is shown in Fig. 3. From Eq. (5) we find

$$\left(\frac{\partial T}{\partial \nu_B}\right)_{S=35\text{‰}, \nu_B=7.5\text{GHz}} = t_1 + St_6 \approx 0.055 \text{ }^\circ\text{C}/\text{MHz}. \quad (8)$$

Thus uncertainties in Brillouin shift measurements of 1 MHz would lead to temperature uncertainties of approximately 0.06 °C; such a measurement accu-

Table 3. $n, v_s(\text{m/s}), \nu_B(\text{GHz})$ for the Ranges $30‰ \leq S \leq 40‰$ and $0 \text{ }^\circ\text{C} \leq T \leq 40 \text{ }^\circ\text{C}$, with $p = 0$ atm and $\lambda = 532.57$ nm

S	30			32.5			35.0			37.5			40.0		
	n	v_s	ν_B	n	v_s	ν_B	n	v_s	ν_B	n	v_s	ν_B	n	v_s	ν_B
0	1.34197	1442.39	7.26907	1.34247	1445.74	7.28863	1.34296	1449.08	7.30821	1.34346	1452.43	7.32781	1.34396	1455.78	7.34744
5	1.34173	1464.26	7.37796	1.34222	1467.45	7.39672	1.34270	1470.64	7.41550	1.34319	1473.84	7.43430	1.34368	1477.03	7.45312
10	1.34142	1483.68	7.47408	1.34189	1486.73	7.49210	1.34237	1489.78	7.51014	1.34285	1492.83	7.52821	1.34332	1495.89	7.54628
15	1.34103	1500.82	7.55825	1.34189	1503.74	7.57560	1.34197	1506.67	7.59297	1.34243	1509.59	7.61036	1.34290	1512.51	7.62777
20	1.34156	1515.86	7.63130	1.34150	1518.66	7.64805	1.34149	1521.47	7.66482	1.34195	1524.27	7.68160	1.34241	1527.08	7.69840
25	1.34002	1528.95	7.69407	1.34048	1531.64	7.71029	1.34093	1534.34	7.72652	1.34139	1537.05	7.74277	1.34185	1539.75	7.75903
30	1.33940	1540.26	7.74740	1.34985	1542.86	7.76314	1.34031	1545.47	7.77891	1.34077	1548.08	7.79469	1.34122	1550.69	7.81049
35	1.33870	1549.95	7.79211	1.33916	1552.48	7.80746	1.33961	1555.00	7.82283	1.34007	1557.53	7.83821	1.34052	1560.06	7.85361
40	1.33792	1558.21	7.82906	1.33838	1560.66	7.84408	1.33884	1563.12	7.85912	1.33930	1565.58	7.87418	1.33976	1568.04	7.88925

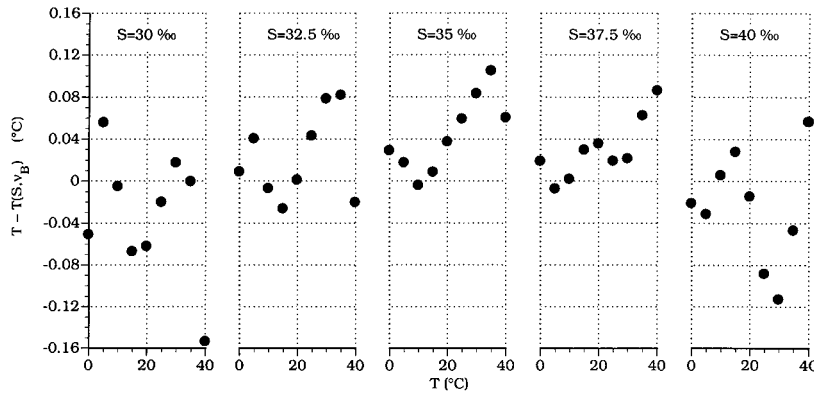


Fig. 1. Difference between the temperature T in Table 3 and $T(S, v_B)$ given by Eq. (5) for each (S, v_B, T) combination listed in Table 3. The abscissa is temperature rather than v_B so as to simplify identification of the corresponding points in Table 3.

racy can be expected with a Brillouin lidar (see Subsection 3.B).

The dependence of sound speed on the salinity for fixed values of v_B is shown in Fig. 5. From Eq. (6) we find

$$\left(\frac{\partial v_s}{\partial S}\right)_{v_B=7.5\text{GHz}} = c'_5 \approx -0.25 \text{ m/s/‰}. \quad (9)$$

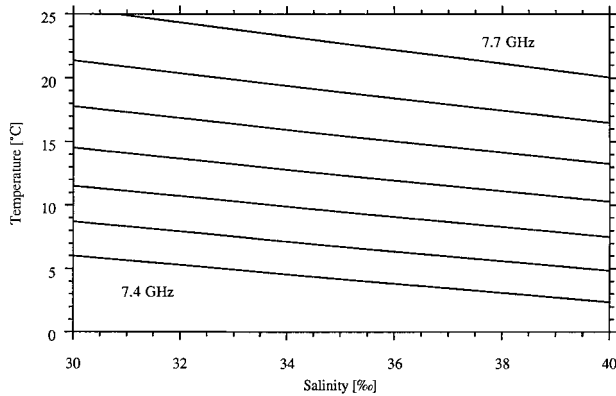


Fig. 2. Temperature $T(S, v_B)$ shown as a function of salinity S with the Brillouin frequency shift v_B as a parameter between 7.4 and 7.7 GHz in steps of 50 MHz.

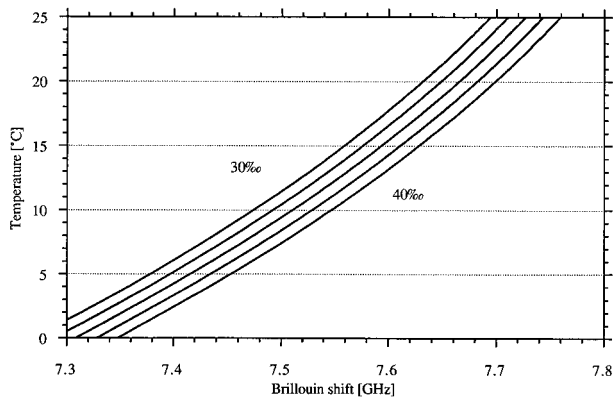


Fig. 3. Temperature $T(S, v_B)$ shown as a function of the Brillouin frequency shift v_B for values of salinity S from 30‰ to 40‰ in steps of 2.5‰.

Thus an uncertainty in the salinity of 0.5‰ gives an uncertainty in the sound speed of ≈ 0.12 m/s. Actually, a rough knowledge of the salinity (a few parts per thousand) is sufficient to determine the sound speed with an uncertainty less than 1 m/s.

The dependence of v_s on v_B is shown in Fig. 6. From Eq. (6) we find

$$\left(\frac{\partial v_s}{\partial v_B}\right)_{S=35\text{‰}, v_B=7.5\text{GHz}} = c'_1 \approx 0.2 \text{ m/s/MHz}. \quad (10)$$

Thus a measurement error in the Brillouin shift of 1 MHz corresponds to an uncertainty of 0.2 m/s for the sound speed.

To summarize these results, use of the three relations [Eqs. (1), (2), and (4)] has enabled us to rephrase the problem so that only the uncertainties in v_B and S are relevant. For example, temperature has become a derived quantity, and its uncertainty is irrelevant to the determination of the uncertainty in v_s .

3. Measurement Uncertainty in the Temperature and Sound Speed

The root-mean-square error (uncertainty) in measurements of the temperature δT and sound speed δv_s can be expressed in terms of the uncertainties δS and δv_B in the independent parameters S and v_B :

$$\delta T = \left[\left(\frac{\partial T}{\partial S}\right)^2 (\delta S)^2 + \left(\frac{\partial T}{\partial v_B}\right)^2 (\delta v_B)^2 \right]^{1/2}, \quad (11)$$

$$\delta v_s = \left[\left(\frac{\partial v_s}{\partial S}\right)^2 (\delta S)^2 + \left(\frac{\partial v_s}{\partial v_B}\right)^2 (\delta v_B)^2 \right]^{1/2}. \quad (12)$$

The following two subsections address estimates for δS and δv_B .

Table 5. Coefficients in the Empirical Expression for $v_s(S, v_B)$ at 532.57 nm

$c'_0 = 1496.3$	$c'_1 = 203$
$c'_2 = 19$	$c'_3 = 78$
$c'_4 = 190$	$c'_5 = -0.2453$
$c'_6 = -0.254$	$c'_7 = -2.17$

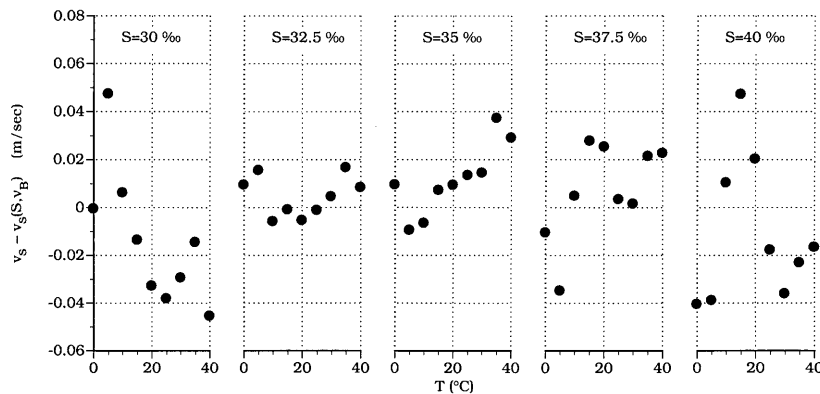


Fig. 4. Difference between the sound speed v_s in Table 3 and $v_s(S, v_B)$ given by Eq. (6) for each (S, v_B, v_s) combination listed in Table 3. The abscissa is temperature rather than v_s so as to simplify identification of the corresponding points in Table 3.

A. Variability of the Salinity in the Ocean

We estimated a standard deviation δS for the salinity by referring to the "Oceanographic Station Profile Time Series."¹⁸ It provides a compilation of oceanographic data obtained from repetitive samples taken along various ocean sections and at fixed coordinates (oceanic weather stations) between 1900 and 1990. We analyzed this data set for the salinity variability at 113 geographical locations distributed over all the

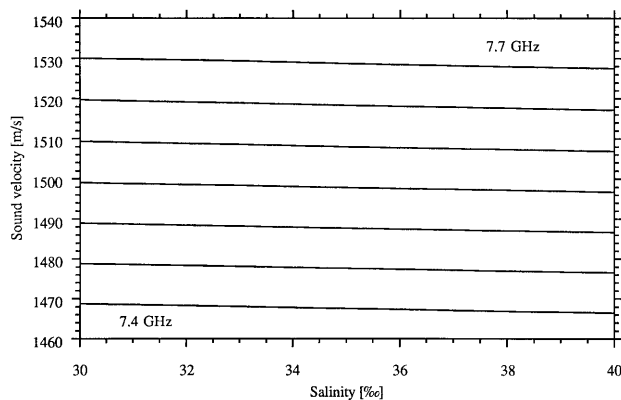


Fig. 5. Sound speed $v_s(S, v_B)$ shown as a function of salinity S with the Brillouin frequency shift v_B as a parameter (in steps of 50 MHz).

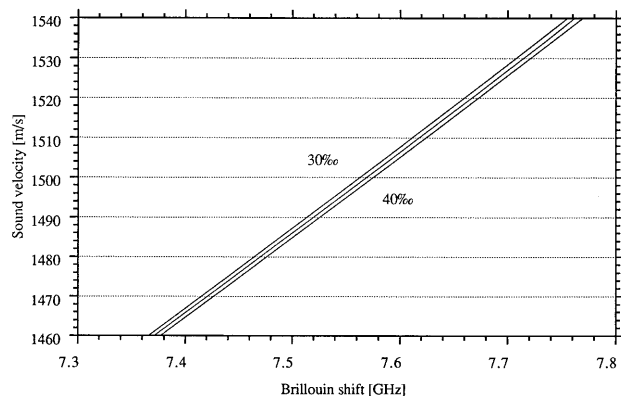


Fig. 6. Sound speed $v_s(S, v_B)$ shown as a function of the Brillouin frequency shift v_B for values of salinity S from 30‰ to 40‰ in steps of 5‰.

oceans of the Earth (more than 75,000 measurements). Each location corresponds to an area of approximately $500 \times 500 \text{ km}^2$. For each location and depth, we computed the mean value and the standard deviation of the observations. Figure 7 shows a typical example located at $32^\circ 15' \text{ S } 17^\circ 30' \text{ E}$ near the Cape of Good Hope at a depth of 10 m and in an area of $30 \times 100 \text{ km}^2$. Over a period of 25 years, all the observed values of salinity lie within a narrow bandwidth having an average value of 34.99‰ with a standard deviation of 0.23‰.

The results of our salinity analysis are summarized in Fig. 8, where the cumulative distribution of the salinity standard deviation is shown for depths of 10 m [Fig. 8(a)] and 100 m [Fig. 8(b)]. It is expressed as a fraction of the total number of locations. In particular, for a depth of 10 m, 50% of the locations have a salinity standard deviation δS less than 0.5‰, and in 92% of the cases, δS is less than 1‰. For a depth of 100 m, δS is less than 0.2‰ for 43% of the locations and is less than 1‰ for 99% of the locations. As expected, the standard deviation generally decreases with depth.

Some locations exhibit seasonal variations, which could be taken into account in the reduction of δS . An example is shown in Fig. 9 for measurements performed in the middle of the North Pacific ($34^\circ 00'$

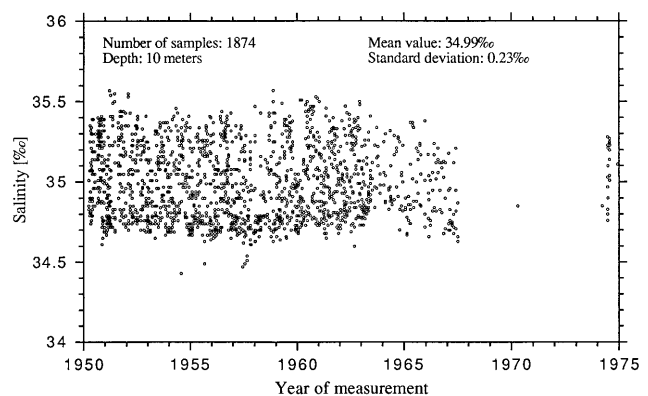


Fig. 7. Observed values of the salinity at $32^\circ 15' \text{ S } 17^\circ 30' \text{ E}$ in the South Atlantic near the Cape of Good Hope at a depth of 10 m. The sampling area is $30 \times 100 \text{ km}^2$.

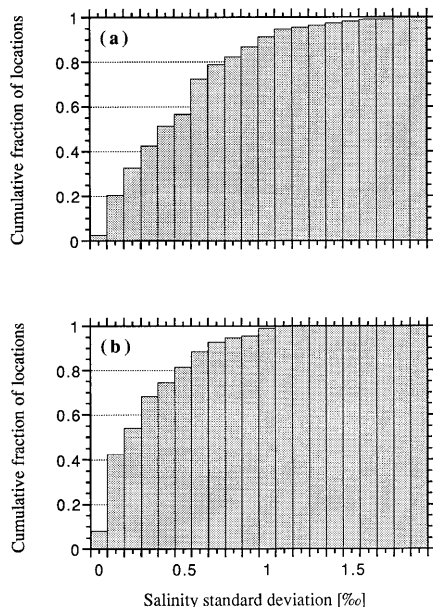


Fig. 8. Cumulative histogram of the variability of the salinity standard deviation for 113 locations for depths of (a) 10 m and (b) 100 m.

N 164°00' E) at depths of 10, 50, and 100 m in an area of $50 \times 50 \text{ km}^2$. For this location, δS would be significantly smaller if the seasonal trend is considered.

We also noticed that, for a given station, δS does not change substantially when increasing or decreasing the horizontal sampling area or when increasing the thickness of the water cell considered at a given depth. This shows that the salinity is fairly stable, and that historical data provide useful salinity data

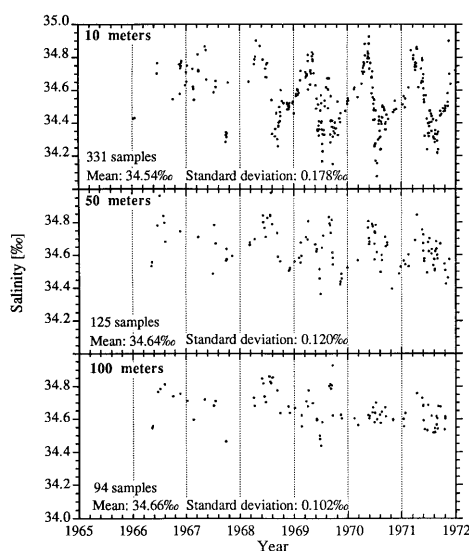


Fig. 9. Seasonal variability of the salinity at 34°00' N 164°00' E in the middle of the North Pacific at depths of 10, 50, and 100 m. Sampling area is $50 \times 50 \text{ km}^2$. The standard deviation represents the overall variability; the seasonal trend is not taken into account in its determination.

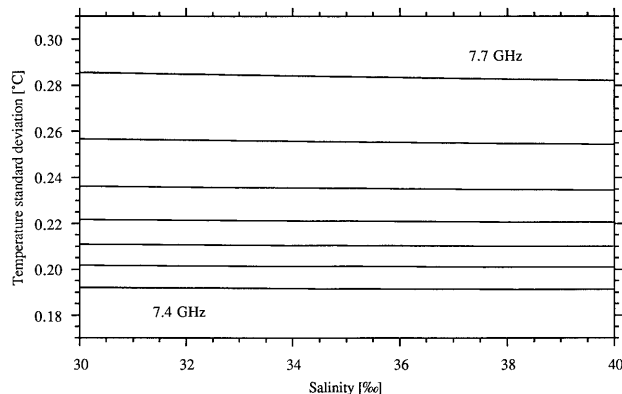


Fig. 10. Uncertainty δT in the determination of temperature as a function of the salinity for $\delta S = 0.5\text{‰}$ and $\delta \nu_B = 1 \text{ MHz}$ and various Brillouin shifts from 7.4 to 7.7 GHz in steps of 50 MHz.

that can be used to determine T and ν_s from Brillouin shift measurements.

B. Brillouin Shift Measurement Accuracy

The use of injection locking techniques enable one to produce stable laser pulses having a Fourier transform limited bandwidth of approximately 50 MHz at a wavelength of 532 nm.^{19,20} The width of the Brillouin line is $\approx 500 \text{ MHz}$, so the linewidth of the pulsed laser as well as its stability are no longer a technical limitation for measurements of the Brillouin shift.

For a range-resolved measurement, a nonscanning high-resolution spectroscopic technique is needed. The so-called edge technique¹³ has been demonstrated to be a powerful method for the measurement of small-frequency changes in an atmospheric Doppler lidar application; wind velocity has been determined with an uncertainty of 0.25 m/s,²¹ corresponding to a frequency change of less than 470 kHz at 1064 nm. This is currently the most sensitive method to be demonstrated for small-frequency shift lidar measurements.²² Such a technique has been considered for oceanic Brillouin shift measurements with a laser wavelength of 532 nm.^{2,9,23}

The statistical uncertainty in the Brillouin shift measurement is the factor limiting the accuracy. This uncertainty depends on laser intensity, laser beam attenuation in the water, depth resolution, solid angles, etc. Based on recent studies,^{2,23} we consider two typical values for the uncertainty in the Brillouin shift measurement; specifically, we take $\delta \nu_B = 1 \text{ MHz}$ (depths or ranges to 75 m) and $\delta \nu_B = 4 \text{ MHz}$ (depths or ranges to 100 m) in the following estimates.

C. Temperature and Sound Speed Measurement Accuracy

Equations (5) and (6) and the estimates of δS and $\delta \nu_B$ made in Subsections 3.A and 3.B allow the evaluation of the temperature and sound speed measurement errors given by Eqs. (11) and (12).

Figure 10 shows the standard deviation in the temperature δT for $\delta S = 0.5\text{‰}$ and $\delta \nu_B = 1 \text{ MHz}$. This

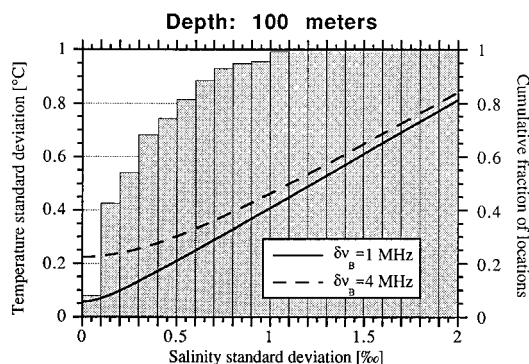


Fig. 11. Uncertainty δT (left side, y axis) in the determination of temperature is plotted as a function of the salinity standard deviation for $\delta v_B = 1$ MHz (solid curve) and $\delta v_B = 4$ MHz (dotted curve). In both cases we have taken $v_B = 7.5$ GHz and $S = 35\text{‰}$. It is superposed on the salinity standard deviation cumulative probability distribution at 100 m (right side, y axis) [same as Fig. 8(b)].

uncertainty δT is independent of T ; it depends only on S and v_B ; it is plotted as a function of S for typical oceanic values between 30‰ and 40‰ and with v_B as a parameter from 7.4 to 7.7 GHz in steps of 50 MHz. For a typical value of $v_B = 7.5$ GHz, the temperature resolution is approximately 0.2°C ; this is valid over the entire temperature range being considered here, 0°C to 40°C .

Figure 11 shows δT (left side, y axis) as a function of δS superposed on the cumulative distribution for the salinity standard deviation at a depth of 100 m (right side, y axis) as in Fig. 8(b). For these results we have taken $v_B = 7.5$ GHz and $S = 35\text{‰}$. When δS is smaller than $\approx 0.3\text{‰}$, the uncertainty δv_B is the dominant contribution to δT . If δv_B is less than 1 MHz, the uncertainty in the temperature is less than 0.15°C for more than 50% of the locations. At 99% of the locations $\delta S < 1\text{‰}$; thus, if $\delta v_B = 4$ MHz, δT is less than 0.5°C at these locations.

Figure 12 shows the standard deviation of the sound speed δv_s for $\delta S = 0.5\text{‰}$ and $\delta v_B = 1$ MHz.

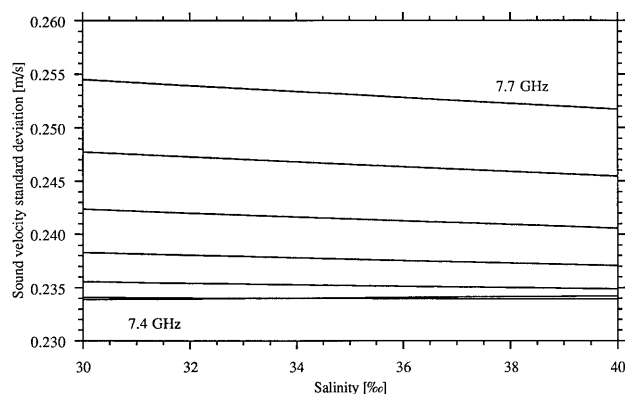


Fig. 12. Uncertainty δv_s in the determination of sound speed as a function of salinity with v_B as a parameter that is varied from 7.4 to 7.7 GHz in steps of 50 MHz. For these calculations we have taken $\delta v_B = 1$ MHz and $\delta S = 0.5\text{‰}$.

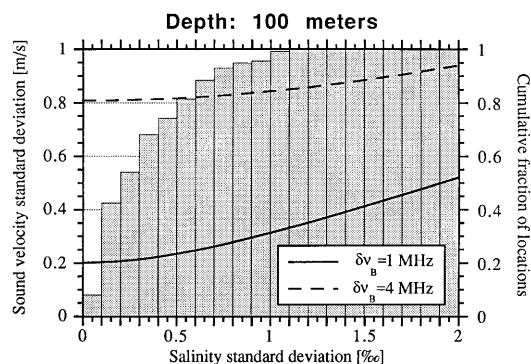


Fig. 13. Uncertainty δv_s (left side, y axis) in the determination of sound speed is plotted as a function of the salinity standard deviation for $\delta v_B = 1$ MHz (solid curve) and $\delta v_B = 4$ MHz (dotted curve). In both cases we have taken $v_B = 7.5$ GHz and $S = 35\text{‰}$. It is superposed on the salinity standard deviation cumulative probability distribution at 100 m (right side, y axis) [same as Fig. 8(b)].

This uncertainty δv_s is independent of v_s ; it depends only on S and v_B ; it is plotted as a function of S for $30\text{‰} \leq S \leq 40\text{‰}$ and $7.4\text{ GHz} \leq v_B \leq 7.7\text{ GHz}$ in steps of 50 MHz. Notice that throughout the range of parameters for Fig. 12, the sound speed uncertainty is always less than approximately 0.25 m/s ; this is valid over the entire sound speed range of ≈ 1440 to $\approx 1570\text{ m/s}$.

Figure 13 shows δv_s (left side, y axis) as a function of δS superposed on the cumulative distribution of the salinity standard deviation (right side, y axis) at a depth of 100 m as in Fig. 8(b). For these results we have also taken $v_B = 7.5$ GHz and $S = 35\text{‰}$. The small slope of $\delta v_s(\delta S)$ is a consequence of the weak dependence of v_s on S . The Brillouin frequency shift is the dominant contribution to the uncertainty in a sound speed measurement. Assuming δv_B is less than 1 MHz, we find that δv_s is less than 0.3 m/s for more than 99% of the locations. Even for $\delta v_B = 4$ MHz, δv_s is still always less than 1 m/s .

In fresh water there are only four parameters since S is already known to be zero. Because there are three relations, measurement of one parameter, v_B , is sufficient to determine the others. For Brillouin lidar measurements in fresh water with $\delta v_B = 1$ MHz, we find $\delta T \approx 0.06^\circ\text{C}$ and $\delta v_s \approx 0.2\text{ m/s}$.

Finally, we note that if one more of the variables v_B , S , T , v_s , and n , or equivalently if one more relation between them is also measured, then all the others, including salinity, can be determined without the necessity of recourse to historical data for salinity. There are several conceivable possibilities, including simultaneous use of Raman scattering, measurements of conductivity, measurements of v_B at two or more wavelengths, and determinations of Brillouin linewidths. For some situations there is also an S - T relationship²⁴ that may be applicable to increase the measurement accuracy.

4. Conclusions

We have obtained rigorous relations between v_B , S , T , v_s , and n and have used them to determine the limits

of accuracy of a Brillouin lidar system. We have also analyzed a large set of historical data to determine the mean and the standard deviation of the salinity. Use of this additional parameter allows us to show that the temperature and sound speed can be retrieved with an accuracy sufficient for many applications. Implementation of a Brillouin lidar system is under way, and the first frequency-resolved measurement of the Brillouin lines with use of a pulsed laser has been recently published.¹⁴

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